

🏠 Kubernetes Lab Challenge: Distributed App with Shared Volume

🎯 Objective

Deploy a **multi-Pod application** on Amazon EKS that uses a **shared volume** (EFS) to read/write data collaboratively.

🧑‍🔬 Scenario

You're building a distributed file-processing app where:

- One pod writes data into a shared folder.
- Other pods consume and process those files.

Use **Amazon EFS** and **EFS CSI Driver** to mount a shared volume across all pods.

🧰 Tools & Services

- AWS EKS
- Amazon EFS
- EFS CSI Driver (via Helm or YAML)
- Kubernetes StatefulSet / Pods

🛠️ Tasks

🟢 Step 1: Provision EFS

- Create an **EFS filesystem** in the same VPC as your EKS cluster
- Create a **mount target** in each subnet used by EKS nodes

📘 Reference:

- <https://docs.aws.amazon.com/efs/latest/ug/creating-using.html>

🟢 Step 2: Install EFS CSI Driver

- Use Helm or apply manifests directly

```
```bash
```

```
helm repo add aws-efs-csi-driver https://kubernetes-sigs.github.io/aws-efs-csi-driver/
```

```
helm install aws-efs-csi-driver aws-efs-csi-driver/aws-efs-csi-driver --namespace kube-system
```

```
```
```

 Reference:

- <https://github.com/kubernetes-sigs/aws-efs-csi-driver>

Step 3: Create Storage Class and PVC

- Create a StorageClass pointing to `efs.csi.aws.com`
- Create a PersistentVolume (PV) and PersistentVolumeClaim (PVC)

Hint: Use `efsVolumeHandle` (EFS FileSystem ID)

 Example:

- <https://docs.aws.amazon.com/eks/latest/userguide/efs-csi.html>

Step 4: Deploy Producer & Consumer Pods

- One pod writes a message/file every 10 seconds
- Another pod lists or processes files from the same volume

 Use `mountPath: /shared-data` in both pods

Step 5: Verify Shared Access

- `kubectl exec` into both pods
- Confirm that data written by one pod is visible to others

Completion Criteria

- EFS file system mounted across all pods
- Producer pod writes files
- Consumer pod processes or lists files successfully

Bonus Challenges

- Use a **StatefulSet** instead of static Pods
- Implement a sidecar to move processed files to S3
- Monitor EFS usage via CloudWatch metrics

📄 Deliverables

Be ready to demonstrate:

- Shared volume mounted using EFS
- Logs showing file creation and processing
- Successful pod-to-volume interaction